

ABDUR REHMAN

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PROFESSIONAL SUMMARY

Motivated Software Engineering student with hands-on experience in object-oriented programming, data structures, and software design principles. Passionate about building scalable, maintainable applications and solving real-world problems through clean, efficient code. Seeking opportunities to apply academic knowledge in a professional environment.

EDUCATION

Bachelor of Science in Software Engineering

Sep 2024 – Ongoing

COMSATS University Islamabad | CGPA: 3.91 / 4.0 | Islamabad, Pakistan

Relevant Coursework: Data Structures & Algorithms, Software Requirements Engineering, Information Security, Database Systems, Software Engineering Concepts, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

Frameworks & Libraries: tkinter, swing, JDBC, STL (C++), pyswip, deap

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, Linux, VS Code, IntelliJ Idea, CLion

Concepts: OOP, Data Structures & Algorithms, Agile/Scrum, CI/CD, SDLC, Requirement Engineering Process

PROJECTS

1. Typing Speed Tester with Multiple Practice Modes

Dec 2025

C++, CMake, STL, Data Structures

- Built a console typing application with 5 practice modes including Challenge Mode with dynamic difficulty and a 3-life system
- Implemented 8 data structures – BST for leaderboards ($O(\log n)$), Graph for weak word recommendation, Hash Map for $O(1)$ lookups
- Tracked per-user WPM, accuracy, and weak key analytics with persistent file-based leaderboard storage

GitHub: https://github.com/abdurehmaan366/Typing_Speed_Tester

2. AI-Enhanced Ride Hailing System

June 2026

Python, Tkinter, CSP, Genetic Algorithm, Prolog, also made an OOP based version in Java

- Implemented three AI techniques – CSP with backtracking and MRV heuristic for driver filtering, Genetic Algorithm for batch ride optimization, and prolog for rule-based decisions
- Engineered dynamic pricing with peak-hour demand multipliers, surge pricing, and premium rider discounts
- Designed OOP architecture across 20 classes covering abstraction, inheritance, composition, polymorphism, and singleton (Initially a complete OOP project on Java)

GitHub: https://github.com/abdurehmaan366/AI_Enhanced_RideHailing

3. Clinical Trial Management System

May 2026

Oracle SQL, Java Swing, JDBC

- Designed a 13-table normalized Oracle database covering full trial lifecycle – enrollment, visits, lab results, adverse events, and audit trail
- Implemented 4 audit triggers, 3 stored procedures, 2 functions, and 4 views (3NF, BCNF, ERD, EERD, joins, roles, transactions)
- Built a Java Swing GUI with live SQL Query Runner, dashboard statistics, and full CRUD across all 13 tables via JDBC

GitHub: https://github.com/abdurehmaan366/Clinical_Trial_Management_System

CERTIFICATIONS & ACHIEVEMENTS

Google Soft-Skills Certificate — PAFLA / Google

Dec 1, 2024

Effective Problem-Solving & Decision-Making — UCI / Coursera

Jan 24, 2025